

Environmental Effects on NDIR-Based CH₄ Monitoring: Characterization and Correction

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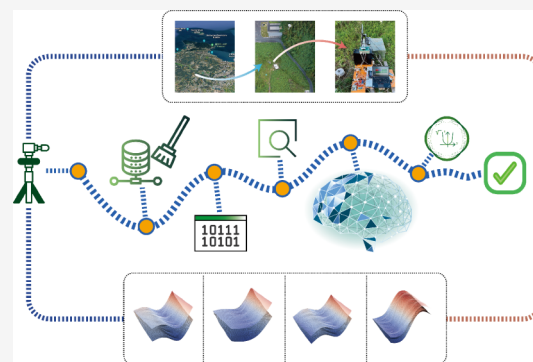
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ABSTRACT: Nondispersive infrared (NDIR) sensors offer high sensitivity, selectivity, and low operational costs, making them particularly well-suited for environmental gas monitoring, where accurate detection of gases such as CH₄ and CO₂ is essential. However, these sensors are highly sensitive to environmental conditions, including temperature and humidity, which can significantly affect detection accuracy. This study characterizes the effects of these conditions and applies machine learning models to correct signal biases caused by multiple environmental factors. Experiments simulating natural environmental conditions for CH₄ monitoring were conducted in the laboratory across a temperature range of 10–40 °C, relative humidity levels of 10–70%, and CO₂ concentrations ranging from 0 to 1000 ppm, revealing significant signal variability under these conditions. The simulations and their results were comprehensively validated at the Ito Natural Analogue Site (INAS), a real-world field-testing location dedicated to investigating environmental impacts. Using machine learning regression algorithms for comprehensive compensation of environmental influences, we successfully mitigated signal biases caused by environmental factors. This offers a cost-effective solution for improving detection accuracy and reliability while reducing system complexity and operational costs.

KEYWORDS: NDIR sensors, CH₄ monitoring, gas detection accuracy, environmental effects characterization, machine learning-based correction



INTRODUCTION

Environmental fluctuations significantly challenge gas monitoring, particularly when measuring trace gases in the atmosphere.^{1–3} This challenge is compounded for NDIR sensors, which, despite their high sensitivity and selectivity for specific gas molecules, are highly susceptible to dynamic environmental conditions.^{4,5} However, their low cost and easy-to-maintain structures also provide broad prospects for their application under such conditions. Compared to the more established NDIR technology for CO₂ monitoring, CH₄ monitoring presents additional challenges due to its significantly lower atmospheric concentration, weaker infrared absorption bands, and higher susceptibility to cross-interference from other gases.^{6–8} Therefore, accurately analyzing the response characteristics of NDIR technology to CH₄ under various environmental conditions and exploring methods to enhance its monitoring capabilities are of significant scientific and practical importance.

To address the challenges posed by environmental fluctuations, recent advancements in NDIR sensor technology have focused on modeling sensor behavior with consideration of ambient temperature, relative humidity, measured responses, and concentration levels as key inputs.^{9,10} Enhancements in environmental corrections and calibration techniques

have substantially improved sensor stability and accuracy, particularly in applications related to automotive and environmental monitoring.^{4,11,12} While these innovations have driven significant advancements in CO₂ sensing, the development of similarly precise and commercially viable NDIR sensors for CH₄ monitoring remains challenging, particularly under dynamic environmental conditions.

For CH₄ monitoring, environmental corrections often require advanced modeling approaches to address the nonlinear impacts of temperature, as well as the need to distinguish CH₄ signals from overlapping absorption features of H₂O and CO₂.^{6,13} The dynamic nature of environmental conditions continues to introduce biases and variability, necessitating the integration of advanced computational methods, such as machine learning and data assimilation, to effectively mitigate these challenges.^{14–19} An NDIR CO₂ sensor with an XGBoost-based drift compensation model

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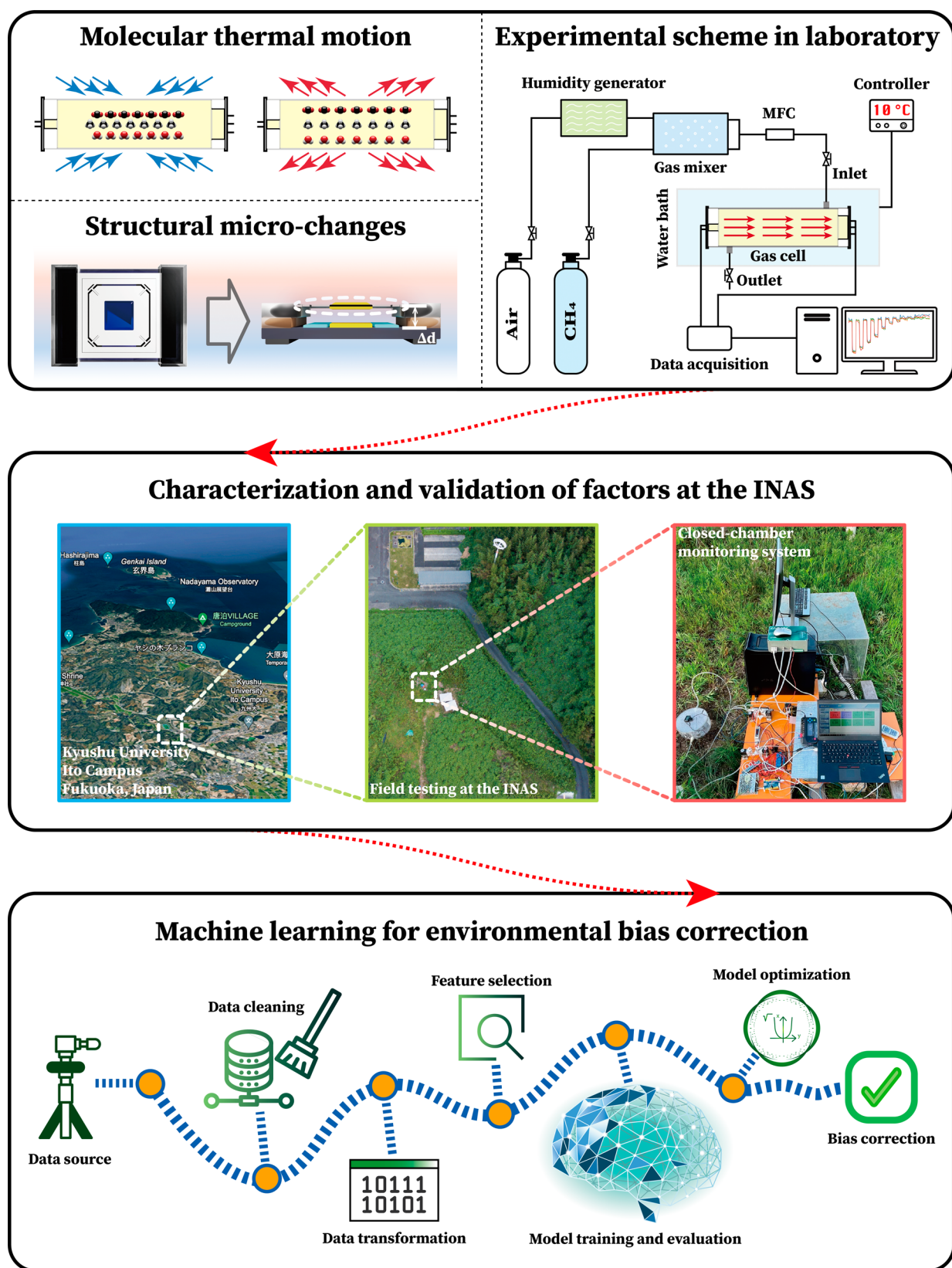


Figure 1. Overview of the study framework, problem definition, laboratory experiments, field-test validation, and bias correction using machine learning techniques.

was evaluated on test and validation datasets, achieving accuracies of 99.83% and 98.90%, respectively, confirming the correction model's effectiveness and robustness.¹⁷ Sim-

ilarly, random forest calibration models have demonstrated significant reductions in mean absolute errors across multiple gas pollutants, including CO, CO₂, NO₂, and O₃, proving their

effectiveness in improving gas monitoring accuracy.¹⁹ These advancements have the potential to significantly enhance the precision, accuracy, and efficiency of NDIR sensor calibration, particularly for low-cost CH₄ monitoring systems.

In this study, we examine the impact of environmental fluctuations on CH₄ monitoring using NDIR technology, integrating laboratory simulations with field-test validation. We focus on achieving accurate measurement and correction under dynamic environmental conditions through machine learning techniques. To this end, we conduct a comparative analysis of six machine learning models, enhanced with Bayesian optimization, to correct measurement biases introduced by environmental variability. Our approach significantly improves the robustness and precision of CH₄ concentration measurements, providing a reliable solution for real-world applications.

MATERIALS AND METHODS

To characterize the key factors affecting NDIR-based CH₄ monitoring, this study integrates both laboratory-based simulations and real-world field tests to analyze the impact of environmental fluctuations. Based on these findings, a robust correction method is developed to mitigate the effects of sensor signal biases caused by environmental changes. An overview of the study framework, including problem statements, methods, key components, and relationships, is presented in Figure 1. The process begins with initial indoor experiments aimed at characterizing the primary challenges and key environmental factors influencing CH₄ measurements. Comprehensive field tests at the INAS then follow, involving extended measurements to further investigate the effects of outdoor environmental fluctuations under real-world conditions. Finally, machine learning models, optimized with Bayesian techniques,^{20,21} are applied to correct the measurement biases, significantly improving accuracy and reliability in CH₄ monitoring.

Problem Statement. Temperature shifts can introduce variations in gas concentration readings, as NDIR sensors rely on infrared light absorption, which is influenced by changes in air density and humidity.^{2,22,23} These fluctuations affect gas molecule density, condensation, and the distribution of molecules during heating and cooling.²⁴ Furthermore, humidity variations complicate infrared absorption, while thermal expansion and contraction in sensor components cause drift, collectively impacting measurement performance.

Laboratory experiments under controlled conditions and relatively short durations provide a stable context for observing these effects and isolating specific sources of variability. However, in real-world applications, environmental conditions are rarely constant. Variations in soil properties, such as temperature, moisture, salinity, and pH, as well as atmospheric factors including ambient temperature and CO₂ concentration, create dynamic environments where these factors interact in unpredictable ways.^{25,26} Seasonal temperature fluctuations and regional differences further exacerbate these challenges, complicating the accurate monitoring of CH₄ emissions in diverse settings.

Based on observations from both laboratory experiments and field tests, we define bias as the comprehensive deviation observed in field tests, calculated by subtracting the sensor drift (determined under controlled laboratory conditions) from the measurements. This bias is solely attributable to environmental fluctuations, such as soil properties and atmospheric conditions, which affect the sensor's performance under

dynamic and fluctuating environmental conditions. Characterizing these biases and the conditions contributing to them is crucial for enhancing the accuracy of CH₄ monitoring and addressing the complexities introduced by dynamic environmental interactions.

Environmental Simulation for Laboratory Experiments. To accurately replicate the influence of environmental fluctuations on CH₄ measurements, laboratory experiments were conducted using controlled setups designed to simulate variations in temperature and humidity. These setups, including water baths and humidity generators,^{27–29} allowed us to observe the sensor's response to changing environmental conditions. The water bath was particularly instrumental in simulating natural temperature variations, enabling precise control over the gas cell's exposure to different temperatures, while the humidity generator was essential for maintaining stable moisture levels, allowing for independent regulation of environmental factors.

The experimental process, as depicted in Figure 1, involved testing the sensor's performance across a range of conditions: temperatures from 10 to 40 °C (in 10 °C increments) and relative humidity from 10% to 70% (in 20% increments). Additionally, a comparative analysis was conducted with varying mixed CO₂ concentrations (0 to 1000 ppm in 500 ppm increments), using pure N₂ serving as a control. This resulted in a total of 64 distinct combinations, simulating real-world environmental scenarios. Detailed methodologies, including the measurement point layout and sensor positioning, are provided in Figures S1 and S2. These controlled experiments established a benchmark for assessing the influence of environmental factors on CH₄ detection and guided the development of subsequent field tests to further explore these effects under real-world conditions.

Field Testing at the INAS. To investigate the influence of environmental fluctuations, including atmospheric and soil variations, on CH₄ measurements, we conducted a baseline survey at the INAS site on the Ito Campus of Kyushu University in Fukuoka City, Japan. The site is located at an elevation of 74 m and features a humid subtropical climate with mild winters, hot summers, and significant summer rainfall. In 2023, the mean annual temperature was 17.6 °C, with extremes ranging from 36.1 to −1.8 °C, and annual precipitation of 1650 mm. Relative humidity averaged around 70%, and the site received approximately 1800 h of sunshine annually. The INAS site's weathered granite soils, moderate temperature range, and stable humidity levels provide a controlled environment for evaluating the performance of NDIR sensors under well-defined conditions.

Previous studies at this site recorded background soil CO₂ fluxes below 0.0187 mol m^{−2} h^{−1} for soil temperatures between 12.8 and 25 °C.³⁰ Surface CO₂ fluxes, measured using the closed-chamber method, ranged from 0.00097 to 0.0186 mol m^{−2} h^{−1}.³¹ However, no CH₄ emissions have been detected in the area, likely due to the absence of wetlands, permafrost, or other typical CH₄ emission sources.^{32–35} The site is primarily covered by grasses and shrubs and lacks the water-saturated soils necessary for substantial CH₄ production.

A detailed schematic of the measurement methods is provided in Figure S3, illustrating the experimental setup for CH₄ monitoring under real-world conditions. The setup features a chamber-based method for collecting soil gas emissions using two NDIR sensors for the simultaneous measurement of CO₂ and CH₄. Instruments were also

deployed to monitor the physical and chemical properties of the soil, ensuring a comprehensive assessment of environmental factors affecting gas measurements.

By establishing a baseline under these stable conditions, this study identifies potential biases arising from atmospheric and soil-related factors, with the granite-derived soils offering a reliable low-emission reference environment. The results confirm the sensors' applicability in settings with minimal organic matter and moderate gas exchange rates, providing a benchmark for exploring sensor performance in more variable environments. This controlled baseline framework supports the refinement of NDIR sensor applications and helps to address biases introduced by diverse environmental conditions.

Machine Learning-Based Bias Correction. Given the complexity and unpredictability of environmental fluctuations, it is inevitable that NDIR sensors will experience measurement biases. Addressing biases from atmospheric variations and soil conditions through traditional technical approaches would require substantial investment and technological advancements, making it highly challenging to fully mitigate their impact. As a result, leveraging machine learning techniques has emerged as a promising alternative to enhance the sensor's performance by predicting and correcting these measurement biases.

In this study, machine learning regression models were employed to predict and correct measurement biases caused by environmental fluctuations using both laboratory and field-test data. The field-test results revealed significant measurement biases compared with the controlled laboratory conditions. The average CH₄ response ratio in air, calculated as shown in eq 1, was determined to be 1.182×10^{-3} , with a negligible variation of $\pm 7.24 \times 10^{-5}$, based on five measurements conducted under ideal laboratory conditions.³⁶ The total bias B was determined using eq 2, representing the bias caused by environmental fluctuations, including atmospheric and soil variations, that were not accounted for under laboratory conditions.

$$R_{\text{ratio}} = \log_{10} \frac{R_{\text{reference}}}{R_{\text{target}}} \quad (1)$$

where $R_{\text{reference}}$ represents the baseline signal response from the reference gas, and R_{target} refers to the signal response of the target gas under various controlled conditions.

$$B = M_{\text{field}} - \bar{M}_{\text{lab}} \quad (2)$$

where M_{field} is the measurement obtained during the field tests, and $\bar{M}_{\text{lab}}$ is the baseline value, calculated as the average of five measurements taken in controlled laboratory conditions.

To address the measurement biases, six machine learning regression models were tested and compared: linear regression, decision tree, random forest, gradient boosting, XGBoost, and LightGBM.^{37–42} These models were trained on field data sets, comprising 305 data points, to improve the accuracy of CH₄ measurements under variable environmental conditions. The rationale for selecting these models lies in their demonstrated efficacy in regression tasks under diverse and complex conditions. Linear regression provides a baseline due to its simplicity and interpretability. Decision tree was selected for its capacity to model nonlinear relationships. Ensemble methods such as random forest and gradient boosting are known for their robustness against overfitting and ability to capture intricate data patterns. XGBoost and LightGBM, as advanced

gradient boosting algorithms, offer superior computational efficiency and competitive accuracy, making them benchmarks in modern machine learning applications.

Bayesian hyperparameter optimization was applied to fine-tune each model for regression prediction.²⁰ Unlike traditional grid or random search methods, Bayesian optimization efficiently explores the parameter space using a probabilistic model of the objective function. This approach balances exploration and exploitation, enabling faster convergence to optimal solutions. Key hyperparameters, such as learning rate, maximum depth, and number of estimators, were defined for each selected model. Gaussian process regression was used as the surrogate model, and the expected improvement acquisition function guided the search process, ensuring efficient use of computational resources.

After the initial model selection, four models underwent hyperparameter optimization, with specific parameter settings provided in Tables S1–S4. The models' performance was evaluated using key metrics such as mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination (R^2), as shown in eqs 3–6. Ultimately, the most robust model for correcting measurement biases was selected based on these performance criteria.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

RESULTS AND DISCUSSION

Basic Analysis of Temperature Sensitivity. The laboratory experimental data set is divided into seven temperature groups, each representing a 5 °C interval, based on the gas detector's temperature recordings. This division highlights the sensor's signal response under varying dry gas temperature conditions, as shown in Table 1. It reveals distinct response trends in pure dry air during "heating" and "cooling" states across these temperature ranges, as illustrated in Figure 2a. The gas detector's response significantly increases with

Table 1. Temperature Ranges with Corresponding Heating, Cooling, and Difference Values

Group	Range (°C)	Ambient (°C)	R_{heating} (mV)	R_{cooling} (mV)	ΔR (mV)
1	(15,20)	12.1 ± 1.2	170.709	169.215	1.494
2	(20,25)	16.6 ± 2.5	172.804	171.508	1.296
3	(25,30)	21.9 ± 3.0	175.253	174.715	0.539
4	(30,35)	27.1 ± 3.2	177.658	177.296	0.362
5	(35,40)	31.8 ± 4.0	180.070	179.789	0.282
6	(40,45)	36.5 ± 5.0	182.415	182.347	0.069
7	(45,50)	40.5 ± 3.6	184.436	184.417	0.019

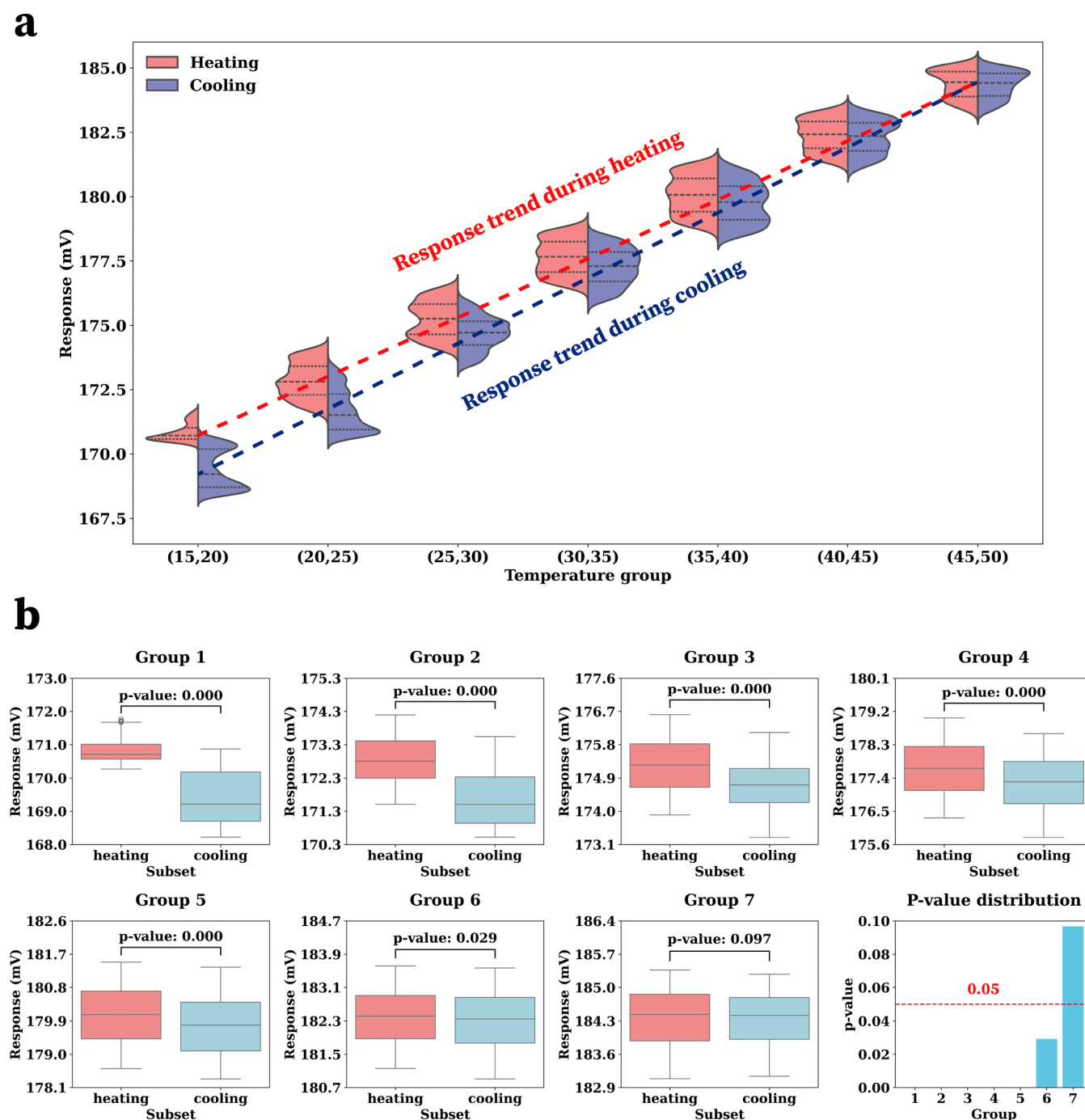


Figure 2. (a) Response trends for each temperature range during the heating and cooling stages. (b) Statistical significance analysis of gas responses for each temperature group.

rising temperature, highlighting the strong influence of temperature on the detector's performance. Moreover, differentiated response trends during the heating and cooling stages are observed, with the differences being most pronounced in lower temperature ranges. The median responses and corresponding temperatures for each temperature range during these two stages were fitted using traditional linear models, as illustrated in Figure S4. Although these models achieve a high R^2 , their generalization capabilities are clearly limited. Neither the heating nor cooling models can accurately capture the relationship between the gas detector's signal response and various temperature states, indicating the need for more robust

regression models that can better handle the complexity and variability of the data.

Further statistical significance analysis, as shown in Figure 2b, reveals the variance in gas responses within each temperature range during "heating" and "cooling" subsets. As the temperature increases, there is a general decrease in variability between the two conditions, suggesting that responses are more consistent at higher temperatures. Notably, only the temperature range 45–50 °C has a p value greater than 0.05, which indicates the most stable signal levels and the least amount of signal variability under higher temperature conditions. This pattern, which exhibits significant variances

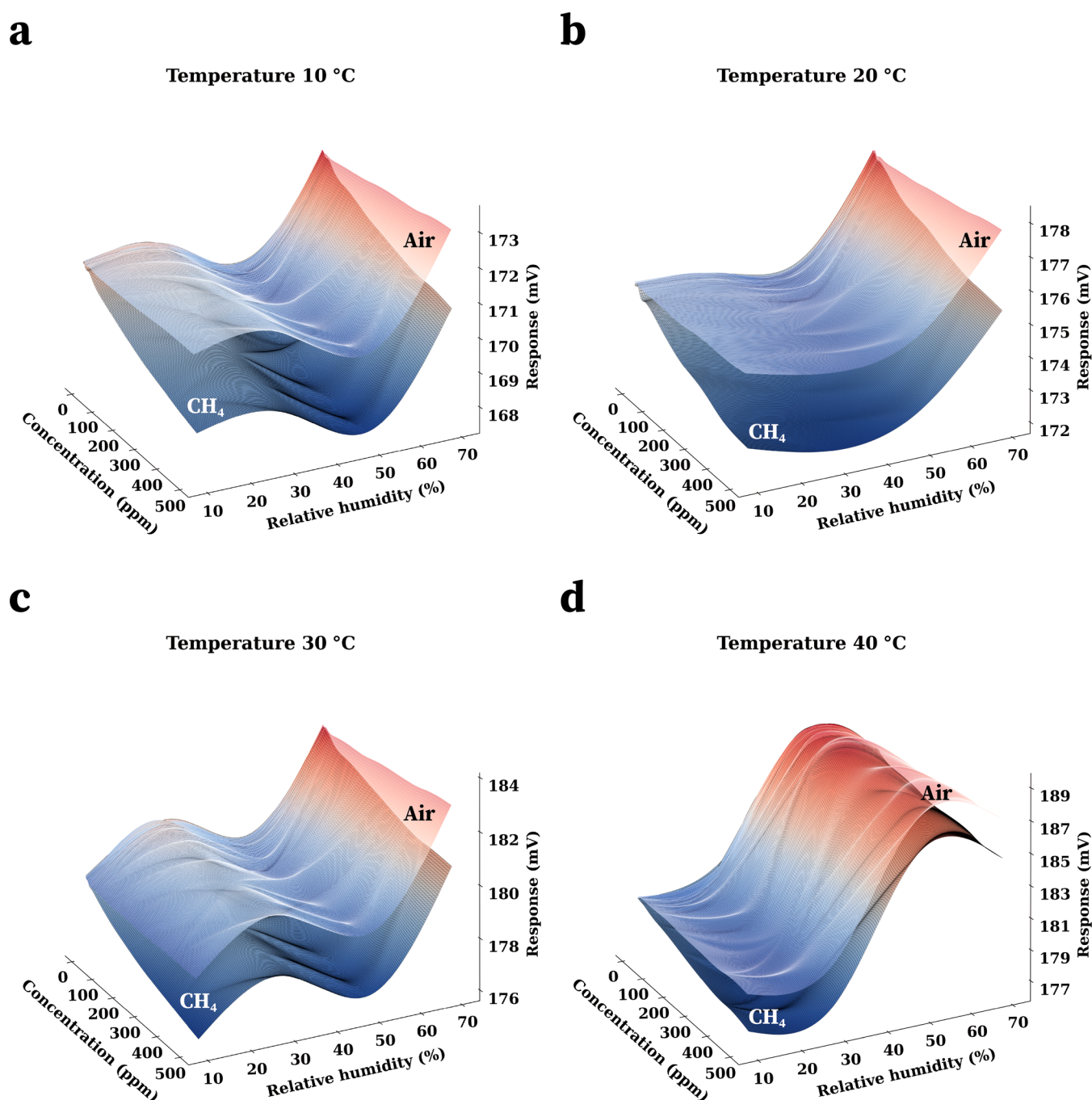


Figure 3. Cross-influence analysis of temperature and humidity on CH₄ concentrations ranging from 5 to 500 ppm, with the color scale representing sensor response variations, where blue indicates lower responses and red indicates higher responses: (a) 10 °C, (b) 20 °C, (c) 30 °C, and (d) 40 °C.

between the temperature increase and fall stages, closely resembles the daily and yearly cycles of temperature fluctuations. Such similarities may help explain the challenges in accurately measuring low daytime fluxes of CH₄ when using the flux-gradient technique.⁴³

These observations are particularly important in practical applications where temperature fluctuations are common, such as in outdoor environments or industrial settings. Stable signal levels at higher temperatures could lead to more reliable measurements, improving the accuracy and consistency of CH₄ detection. High temperatures can also reduce interference from water vapor, a common issue in environmental gas sensing, as

seen in certain commercial NDIR sensors with built-in heating modules that operate around 50 °C. This becomes especially beneficial in outdoor environments, where seasonal and regional temperature variations can cause fluctuations in sensor readings. For instance, colder seasons or high-altitude regions often experience increased water vapor condensation, leading to distortion in CH₄ measurements. Maintaining stability at elevated temperatures helps mitigate these effects. Nonetheless, potential trade-offs, such as greater sensor complexity, higher energy consumption, and increased maintenance demands, must also be carefully evaluated.

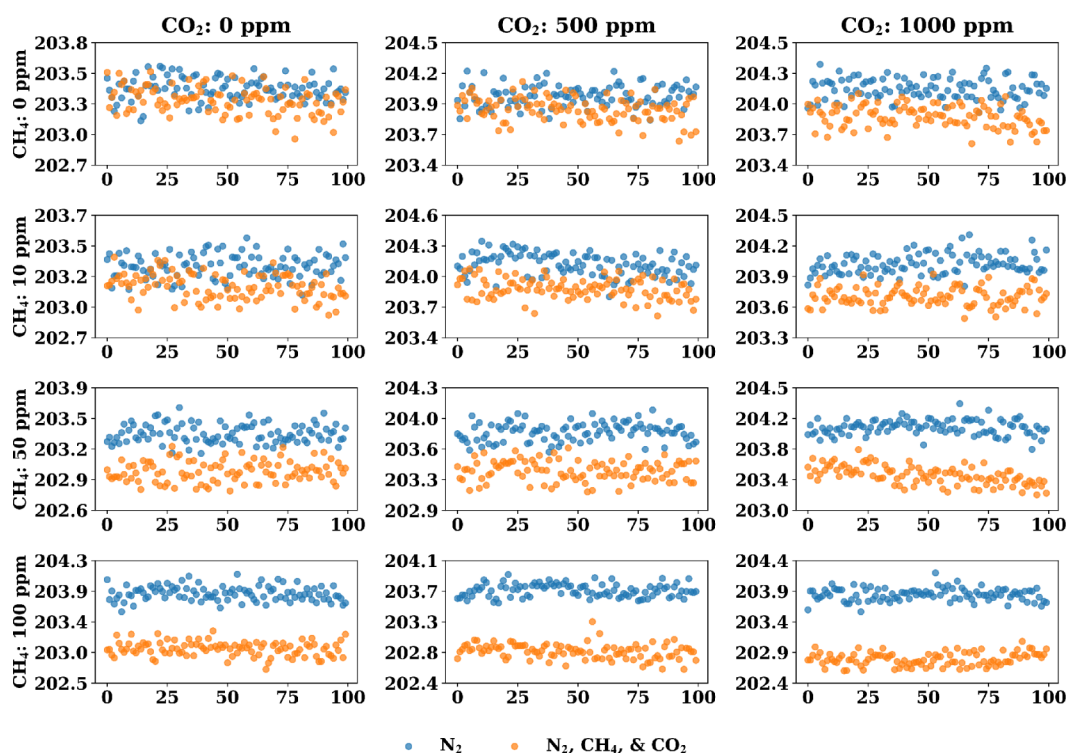


Figure 4. Sensor selectivity analysis under different gas concentrations of CH₄ and CO₂.

Cross-Influence of Temperature and Humidity. Based on simulations using the HITRAN database, which provides high-resolution molecular absorption data, reported in our previous study,³⁶ H₂O molecules are likely a significant factor affecting the NDIR analysis of CH₄ gas. To investigate this, we conducted a cross-influence analysis within an environmental temperature range of 10–40 °C, with increments of 10 °C, and a relative humidity range of 10–70%, with increments of 20%. Under these varying conditions, we observed the response trends of a gas detector for CH₄ concentrations ranging from 5 to 500 ppm. The signal response trends are shown in Figure 3a–d, where the top surface represents baseline reference signals measured from ambient air without added CH₄, and the bottom surface represents variations under different relative humidity conditions for distinct CH₄ concentrations introduced into the experimental setup.

At an ambient temperature of 10 °C, the signal response reaches its minimum at a relative humidity of approximately 50%. As the temperature increases to 20 °C, this minimum shifts to around 30% humidity. When the temperature reaches 30 °C, the minimum signal response occurs at humidities of roughly 10% and 50%. By 40 °C, the minimum temperature is observed at 70% humidity. This phenomenon likely arises from the complex interaction of water vapor molecules with CH₄, including absorption band interference and scattering effects, which modulate the infrared signal received by the detector. The highest response values are consistently seen in high humidity environments, particularly around 70%, while the minimum response values shift with temperature, reflecting the varying influence of H₂O molecules on the sensor's performance.

To assess the response limits in the low concentration range of 5–25 ppm of CH₄, the response ratio between reference and target signals was calculated using eq 1. As shown in Figure S5, at temperatures below 20 °C, the signal stability is

poor across all humidity levels, and resolution between CH₄ concentrations is low. At 30 °C and 10% humidity, the response ratio resolution improves, achieving a detection limit of 5 ppm. However, increasing humidity reduces resolution, raising the detection limit to 15–25 ppm. At 40 °C, excellent resolution and a 5 ppm detection limit are achieved at 10% and 30% humidity, but higher humidity again diminishes resolution, pushing the detection limit to 15–25 ppm. Thus, the combined effect of temperature and humidity significantly impacts the CH₄ detection limit, with optimal performance observed at 40 °C and a humidity below 30%.

Impact Assessment of Gas Selectivity. A comparative analysis using N₂ as a reference gas and varying concentrations of CH₄ and CO₂ as target signals provides a comprehensive assessment of the sensor's performance. This approach evaluates the sensor's selectivity and its ability to detect low CH₄ levels in the presence of fluctuating CO₂, offering insights into its response dynamics under real-world conditions. The results, shown in Figure 4, present experimental configurations where CH₄ concentrations were set at 0, 10, 50, and 100 ppm, while CO₂ concentrations were adjusted to 0, 500, and 1000 ppm, with the *x*-axis representing 100 consecutive sampling points and the *y*-axis representing the sensor response in each subplot.

The results reveal that when both CH₄ and CO₂ concentrations are 0 ppm, the reference and target signals exhibit strong alignment, indicating minimal baseline interference. As CH₄ concentrations increase to 10 ppm, a subtle but distinguishable separation emerges between the signals, which becomes progressively pronounced at 50 and 100 ppm. For CO₂ concentrations of up to 500 ppm, the signal trends remain largely stable, with minimal deviations from the patterns observed at lower CO₂ levels. However, at 1000 ppm of CO₂, clear stratification is observed, particularly in the absence of CH₄, resembling the signal differences caused by 10 ppm of

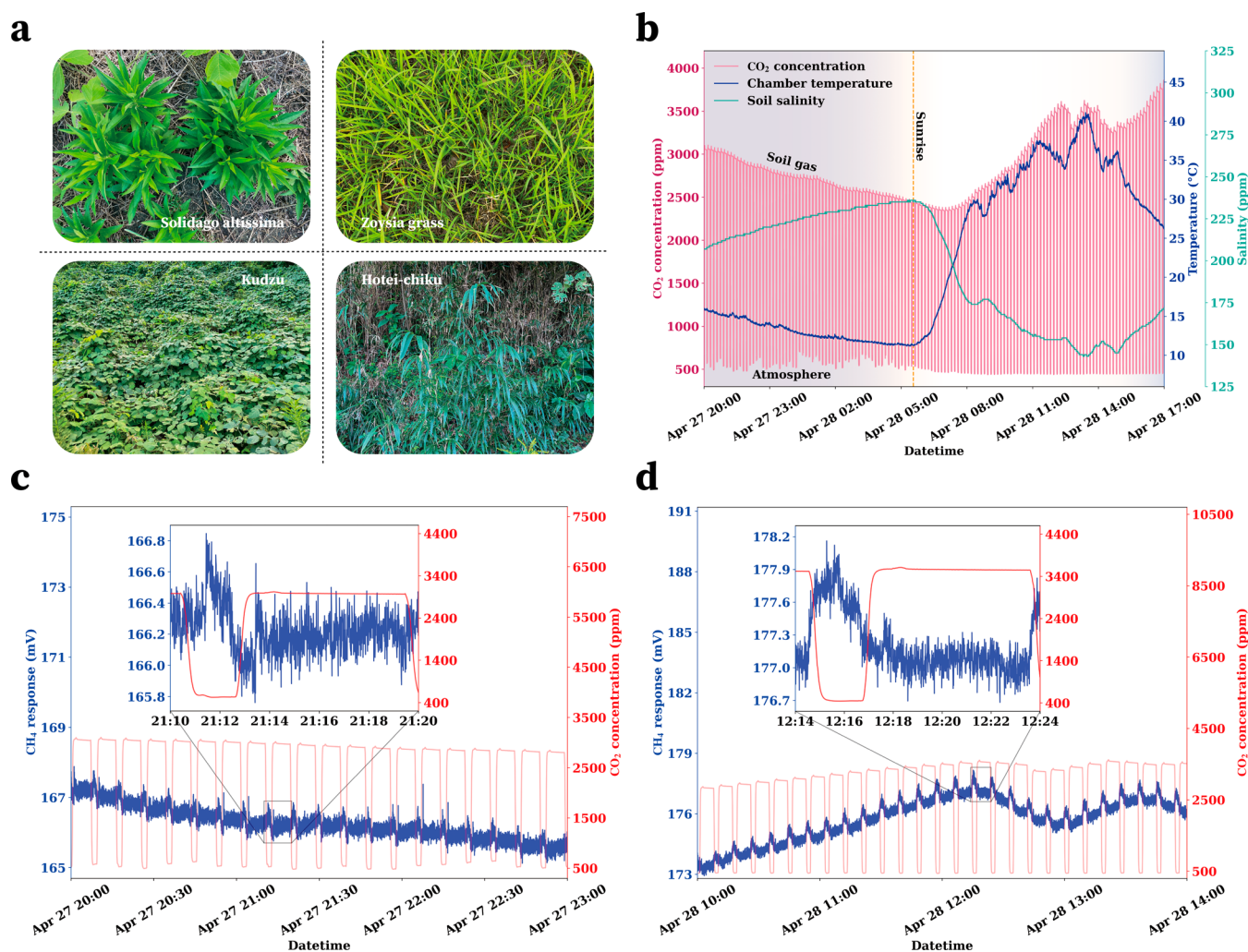


Figure 5. (a) Vegetation coverage around the measurement site, (b) CO₂ concentration and chamber temperature variations over 21 h during the field test, (c) CH₄ response to CO₂ concentration during a 3-h nighttime field test, and (d) CH₄ response to CO₂ concentration during a 4-h daytime field test.

CH₄ at lower CO₂ concentrations. This suggests a non-negligible interference effect, likely arising from overlapping absorption bands between CH₄ and CO₂. A more detailed quantitative analysis of the response ratios for CH₄ and CO₂ mixed gases relative to N₂ is provided in Figure S6, further corroborating these findings.

Higher CO₂ concentrations amplify the CH₄ response ratio due to overlapping infrared absorption bands, as both gases interact with radiation in similar spectral regions. This phenomenon is further intensified by the significant concentration disparity between CH₄ and CO₂: while ambient CH₄ typically remains at a few ppm, CO₂ levels often range from several hundred to thousands of ppm, depending on environmental conditions. Such a difference in magnitude makes CH₄ detection particularly sensitive to CO₂-induced interference, especially in high-CO₂ environments such as urban centers or agricultural regions with substantial biological activity.

Field Test Insights on CH₄ Baseline Dynamics. During a 21-h field test conducted on April 27–28, 2024, at the INAS site, we carried out a comprehensive baseline survey to better understand the local environmental conditions. This baseline survey, conducted at a site with no prior reports of CH₄ emissions, provided an ideal setting for characterizing the

influence of real-world environmental factors on CH₄ monitoring and was critical for interpreting the sensor data. The lack of CH₄ emissions can likely be attributed to the absence of wetlands or permafrost in the area, and the vegetation within 100 m of the test point, primarily consisting of *Solidago altissima*, *Zoysia grass*, Kudzu, and Hotei-chiku, has not been associated with significant CH₄ production in previous studies, as shown in Figure 5a. Moreover, the physical and chemical properties of the soil were also investigated, with a detailed explanation provided in Table S5.

The observed variations in CO₂ concentration, chamber temperature, and soil salinity reveal a strong positive correlation between CO₂ and temperature, while CO₂ shows a negative correlation with salinity, as shown in Figure 5b. A notable finding was that changes in CO₂ concentration lagged behind temperature fluctuations by approximately 2 h, likely due to delayed responses from plant photosynthesis and soil microbial activity. Nighttime showed more pronounced CO₂ fluctuations compared to those of daytime, indicating increased plant respiration emissions during the night. These findings underscore the dynamic nature of CO₂ concentrations in field conditions, which is crucial for understanding their potential impact on CH₄ detection, especially when compared to

controlled laboratory environments where such variability is limited.

To further investigate the CH₄ response, we analyzed two intervals: one at night (20:00–23:00 on April 27) and one during the day (10:00–14:00 on April 28). Figure 5c highlights the nighttime CH₄ response, showing minimal fluctuations with a CO₂ median of 3004.6 ppm. In contrast, Figure 5d shows the daytime CH₄ response with more noticeable variations and a CO₂ median of 3454.4 ppm. This suggests that increased daytime temperature and sunlight enhance biological activity, promoting CO₂ emission variability and leading to greater interference with CH₄ detection via NDIR absorption. The comparison of the CO₂ concentration and CH₄ response characteristics during both periods is summarized in Table 2, illustrating clear differences and trends.

Table 2. Comparison of the CO₂ Concentration and CH₄ Response Characteristics during Nighttime and Daytime Intervals

Interval	CO ₂ Concentration (ppm)		CH ₄ Response		
	Median	Std	R _{air} (mV)	R _{soil} (mV)	R _{ratio}
Nighttime	3004.6	22.0	166.479	166.163	0.0008
Daytime	3454.4	230.9	177.721	177.188	0.0013

Our analysis shows that CH₄ sensor response fluctuations increase as the CO₂ concentration variability grows, particularly when deviations exceed ± 500 ppm from the benchmark. This emphasizes the intricate relationship among temperature, CO₂ emissions, and CH₄ detection. While the field test provided valuable insights, the lack of an internal humidity sensor in the CH₄ detector limits the comprehensiveness of the analysis. Nonetheless, it is likely that humidity, in combination with temperature, plays an important role in influencing the CH₄ response. These findings highlight the limitations of laboratory studies, where controlled conditions may not fully account for the complex interactions of environmental factors in real-world settings.

Bias Correction Accuracy of Machine Learning Models. After the initial model selection, further Bayesian optimization was applied to the most effective models for bias regression predictions using the field-test data set, enabling accurate correction for real-world NDIR-based CH₄ monitoring. Figure 6a shows the actual vs predicted values for each model, along with performance metrics, including MSE, RMSE, MAE, and R^2 . Random forest, gradient boosting, and LightGBM demonstrated the best performance, achieving the lowest MSE and MAE, with R^2 values close to 0.98. Among the models tested, LightGBM demonstrated superior performance by maintaining closely aligned training and cross-validation scores and effectively controlling overfitting, as shown in Figure S7. Random forest and gradient boosting also achieved high accuracy, with R^2 values nearing 0.98, though they exhibited overfitting on smaller data sets. In contrast, linear regression and XGBoost underperformed, displaying higher errors and lower R^2 values, making them less suitable for accurate predictions in this context. These results highlight LightGBM's balanced performance, offering both high accuracy and robust generalization, making it particularly effective for correcting sensor biases in challenging real-world environments.

We selected 2-h periods from both nighttime and daytime for analyzing real-world sensor response biases and evaluating prediction performance, as shown in Figure 6b. The figure shows the sensor response bias (represented by the solid blue line) under monitoring conditions without significant CH₄ release alongside the predicted bias (represented by the dashed red line) based on the LightGBM regression model. Ideally, the sensor response bias should form a stable, unvarying baseline. However, the observed data reveal distinct fluctuations, with alternating peaks and troughs corresponding to environmental changes, particularly temperature variations. These findings reaffirm that environmental factors introduce a notable bias in NDIR-based CH₄ sensor signals. Despite these fluctuations, it is encouraging that the bias can be accurately predicted. Statistical evaluation supports the strong alignment between predicted and actual values, with a t -statistic of 0.64 (p value = 0.52), indicating no significant difference between them. The excellent 95% confidence interval (CI) underscores the robustness of the prediction model, demonstrating its ability to correct for these environmental biases effectively, paving the way for the real-world deployment of reliable CH₄ monitoring solutions.

To better understand how environmental factors influence CH₄ detection and to ensure accurate model predictions, we performed SHAP (SHapley Additive exPlanations) analysis based on the LightGBM model. SHAP is a powerful tool for interpreting machine learning models by quantifying the contribution of each input feature to the model's output.⁴⁴ As shown in Figure 6c, our SHAP analysis reveals that environmental variables such as soil pH, ambient temperature, CH₄ detector temperature, and chamber temperature have a significant impact on model predictions, as indicated by their wider SHAP value distributions. In contrast, variables such as soil salinity and moisture exhibit a lesser effect. This analysis highlights the critical role of temperature, both ambient and sensor-related, in determining CH₄ detection accuracy, providing valuable insights into how environmental factors drive model performance.

Environmental Implications. For NDIR-based CH₄ monitoring, both laboratory and field-test results consistently demonstrate that environmental variations introduce biases, such as fluctuations driven by temperature, humidity, interfering gases, and other environmental conditions. Addressing these biases through improvements in sensor performance presents significant challenges due to limitations in materials, mechanical design, and monitoring principles. Therefore, by collecting data through both laboratory experiments and field tests focused on the characterization of environmental influences on CH₄ monitoring, we explore a new approach to correcting these biases using machine learning techniques.

In the laboratory-scale experiments, we found that low temperatures, especially below 20 °C, significantly affect the variability of the NDIR-based sensor's response, with distinct differences observed between the heating and cooling phases. This highlights the sensor's sensitivity to thermal fluctuations, which may compromise its accuracy in controlled settings. Additionally, fluctuations in CO₂ concentrations within ± 500 ppm of the benchmark had minimal impact on CH₄ detection accuracy, underscoring the sensor's resilience to minor CO₂ variations.

The CH₄ sensor signal exhibited correlated fluctuations with environmental variations, such as soil pH, ambient temper-

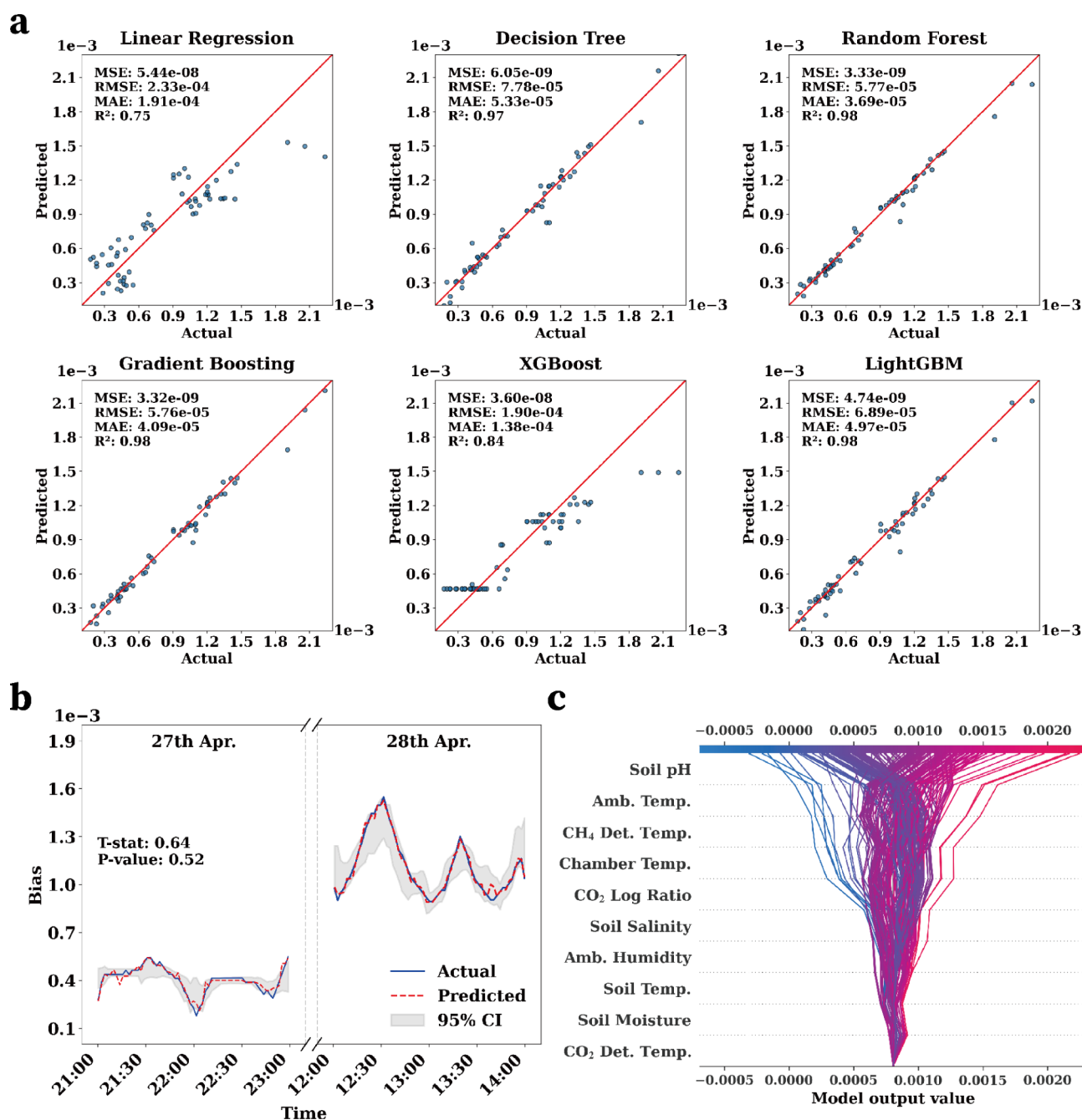


Figure 6. Performance analysis and feature importance for CH₄ detection models: (a) comparison of the performance of different machine learning models, (b) regression predictions based on the LightGBM model, and (c) SHAP analysis highlighting the feature importance of environmental factors in the LightGBM model.

ature, and CO₂ concentration, indicating that the signal is influenced by external conditions. Field tests conducted at INAS confirmed these correlations, with more pronounced fluctuations in the CH₄ signal during daytime conditions compared to the nighttime stability. This suggests that environmental variations, especially those linked to daily temperature fluctuations, exert a stronger influence on the NDIR-based CH₄ sensor performance.

From a broader environmental perspective, the ability to accurately monitor CH₄ emissions under varying conditions is crucial for understanding and mitigating greenhouse gas emissions. Using machine learning regression models, LightGBM demonstrated excellent predictive performance

($R^2 = 0.98$, $MSE = 4.74 \times 10^{-9}$), effectively capturing the underlying trends without overfitting, as evidenced by the learning curves. Additionally, SHAP analysis revealed that soil pH and ambient temperature were the most influential factors, providing a deeper understanding of the model's behavior.

This study presents a cost-effective approach to environmental monitoring by improving detection accuracy while reducing the system complexity and costs. However, further research is required to enhance the real-world applicability. Future work should focus on addressing the confounding effects of temperature variations by considering absolute humidity and investigating the impact of ambient pressure on NDIR performance. Field tests under diverse environ-

mental conditions, such as varying wind speed and precipitation, are crucial to validate the system's robustness. Expanding data sets and refining the system based on these factors will ensure its effectiveness in large-scale environmental monitoring and improve the practical application of the proposed correction methods.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.4c11110>.

Detailed description of the measurement setup and instrument calibration; experimental methodology; additional data processing steps; supplementary figures illustrating key findings; extended discussion on experimental results (PDF)

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Notes

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